
Weekly Practice Assignment

Basic Machine Learning

Topics	Linear Regression, Multiple Linear Regression, Logistic Regression, Naive Bayes
Level	Basic / Beginner
Nature	Weekly Practice Assignment
Platform	Google Colab or Jupyter Notebook

Instructions

- Use **Python** and **Scikit-learn**.
- You may use **Google Colab** or **Jupyter Notebook**.
- Write short comments in your code where necessary.
- Show the output of each important step.
- Submit the completed **.ipynb** notebook.
- Use simple variable names and keep your code properly organized.

Assignment Structure

- Question 1 - Simple Linear Regression
- Question 2 - Multiple Linear Regression
- Question 3 - Logistic Regression
- Question 4 - Naive Bayes Classification

Question 1 - Simple Linear Regression

A student wants to understand whether the **number of study hours** affects the marks obtained in an examination.

Use the following dataset:

Study Hours	Marks
1	35
2	40
3	48
4	55
5	60
6	68
7	72
8	80
9	87
10	92

Tasks

- Create a Pandas DataFrame containing the above data.
- Separate the data into the independent variable **X** and dependent variable **y**.
- Create and train a **Linear Regression** model using Scikit-learn.
- Predict the expected marks of a student who studies for **6.5 hours**.
- Display the **coefficient and intercept** of the trained model.

Question 2 - Multiple Linear Regression

A company wants to predict an employee's **monthly performance score** based on Training Hours, Years of Experience, and Number of Completed Projects.

Use the following dataset:

Training Hours	Experience (Years)	Completed Projects	Performance Score
5	1	2	55
8	2	3	62
10	3	4	68
12	4	5	74
15	5	6	80
18	6	7	85
20	7	8	90
22	8	9	94

Tasks

- Create the dataset using Pandas.
- Define the independent variables and target variable.
- Train a **Multiple Linear Regression** model.
- Predict the performance score for an employee with **Training Hours = 14**, **Experience = 4 years**, and **Completed Projects = 6**.
- Display the coefficients of all three independent variables.

Question 3 - Logistic Regression

A bank wants to predict whether a customer will be **approved for a loan**.

Use the following dataset:

Income	Credit Score	Loan Approved
30000	550	0
35000	580	0
40000	600	0
45000	620	0
50000	650	1
55000	670	1
60000	700	1
65000	720	1
70000	750	1
80000	780	1

0 = Loan Not Approved 1 = Loan Approved

Tasks

- Create the dataset in a Pandas DataFrame.
- Separate the features and target variable.
- Train a **Logistic Regression** classification model.
- Predict whether a customer with **Income = Rs. 48,000** and **Credit Score = 640** will receive a loan.
- Display the **predicted class and prediction probability**.

Question 4 - Naive Bayes Classification

A simple system is required to classify an email as **Spam** or **Not Spam** based on the presence of selected words.

Use the following dataset:

Free	Offer	Meeting	Spam
1	1	0	1
1	1	0	1
1	0	0	1
0	1	0	1
0	0	1	0
0	0	1	0
0	1	1	0
1	0	1	0

1 means the word is present; **0** means the word is absent. **Spam = 1** means Spam; **Spam = 0** means Not Spam.

Tasks

- Create the dataset using Pandas.
- Separate the features **Free**, **Offer**, and **Meeting** from the target **Spam**.
- Train a **Bernoulli Naive Bayes** classifier.
- Predict the class of a new email with **Free = 1**, **Offer = 1**, and **Meeting = 0**.
- Write **2-3 lines** explaining why Naive Bayes is called "**Naive**".

Submission Requirements

- Submit one Google Colab/Jupyter Notebook.
- Create four clearly labelled sections: Simple Linear Regression, Multiple Linear Regression, Logistic Regression, and Naive Bayes.
- Each section should contain the dataset, feature/target preparation, model creation, model training, prediction, and required output.